

Topic, Not Stance:

A Validation Framework for Legal Opinion Similarity Using Circuit Splits

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Abstract

Recent work by [Ganguli et al. \[2024\]](#) demonstrates that the choice of NLP model can produce opposite conclusions about long-run trends in patent similarity, motivating validation-based model selection against domain-specific ground truth tasks. We extend this framework to the legal domain. We construct a hand-curated validation set of 30 federal circuit splits (60 opinions, 12 years, 10 doctrinal areas) and evaluate three embedding models (TF-IDF, Legal-BERT, and GTE) on their ability to distinguish opinions that reach opposite conclusions on the same legal question from opinions that are unrelated. We find that all three models perform near chance: cosine similarity between within-split opposing opinions is statistically indistinguishable from similarity between randomly paired cross-split opinions (Mann-Whitney U, all $p > 0.45$), with TF-IDF most affected by topic-vocabulary overlap (gap of -0.039) and Legal-BERT least affected (gap of $+0.0001$). We interpret this as evidence that current embedding models—including a domain-specific legal model—capture the *topic* of a legal opinion but not its *stance* on the legal question. This finding has implications for any downstream legal NLP application that requires distinguishing legal agreement from legal disagreement (precedent prediction, ideological analysis, doctrinal evolution): text similarity alone is insufficient. We release our validation set and pipeline as a public benchmark.

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1 Introduction

The choice of NLP model is rarely incidental. [Ganguli et al. \[2024\]](#) recently demonstrated that different embedding models (TF-IDF, doc2vec, S-BERT, GTE, PaECTER, OpenAI `text-embedding-3-large`) produce strikingly different and sometimes opposite representations of patent similarity. Applied to the same empirical question of how contemporaneous invention similarity evolved over 1836–2023, the choice of model determines the answer. GTE shows similarity declining for 150 years; TF-IDF shows similarity rising over the same period. Model selection, they argue, is not a technical preliminary but the substantive empirical question.

Their proposed resolution is a four-step pipeline: (1) represent each document as a vector under model m ; (2) compute pairwise cosine similarity; (3) evaluate each candidate model against domain-specific ground truth tasks; (4) select the model with the strongest validation performance. The validation tasks are doing the real work in this framework. [Ganguli et al. \[2024\]](#) use *patent interferences*—USPTO proceedings in which expert examiners determine, on legal grounds, that two patent applications describe substantively the same invention—as their primary ground truth. The intuition is clean: a good similarity model should assign high cosine similarity to pairs that an expert has identified as describing the same idea.

This paper asks whether the same framework can be extended to a different but adjacent domain: federal judicial opinions. Judicial opinions are an attractive target. They are publicly available through the CourtListener archive, structurally consistent (majority, dissent, concurrence), and densely networked through citations. They have generated substantial recent NLP interest, with Legal-BERT [[Chalkidis et al., 2020](#)] emerging as a widely used domain-adapted embedding model. And they pose substantive empirical questions: are dissents becoming more semantically distant from majorities over time? Do circuit courts converge or diverge on contested doctrinal questions? Is there a measurable signature of ideological polarization in the language of legal reasoning?

But applying the validation framework directly turns out to surface a fundamental structural difference between the two domains. To see why, it helps to clarify what a circuit split is and why it serves as a natural ground truth signal for legal disagreement.

The United States federal court system is organized into thirteen regional appellate courts called “circuits,” each with binding authority only within its own geographic territory. The Ninth Circuit covers most of the West Coast; the Second Circuit covers New York and parts of the Northeast; and so on. When a novel legal question arises—say, whether a hyperlink in an employer email is sufficient notice to start the 90-day clock on an employment discrimination lawsuit—different circuits can, and frequently do, reach different conclusions. The First Circuit might say the hyperlink notice is inadequate; the Eighth Circuit might say it is adequate. Both opinions are precedential, both are reasoned at length, both bind the lower courts in their respective territories. Until the Supreme Court intervenes (which happens for only a small fraction of splits), federal law on the same question literally differs depending on where one lives.

This is what legal scholars call a **circuit split**: two or more federal appellate courts reaching

opposite conclusions on the same legal question. For our purposes, a circuit split is an ideal ground truth signal for a specific kind of relationship between two documents—two opinions that (a) address the same legal question, (b) draw on the same body of statutory and constitutional text, (c) cite a heavily overlapping set of prior cases, and (d) arrive at opposite conclusions. The Congressional Research Service catalogues these splits annually for Congress, naming the controlling cases on each side. We use these catalogues as our source.

The patent paper’s ground truth (interferences) is, by contrast, a *topic identity* signal: two patents are “interfering” when they describe the same invention. Cosine similarity in any reasonable embedding space should track this well; “describes the same thing” and “embeds to a similar vector” are operationalizations of the same underlying construct. Circuit splits are structurally different. Circuit-split opinions discuss the same legal question, cite the same precedents, and use the same doctrinal vocabulary. What distinguishes them is not what they are *about*, but where they *land*. Text similarity, as conventionally implemented, is not designed to detect this.

This paper makes three contributions.

First, we adapt the [Ganguli et al. \[2024\]](#) validation framework to the legal domain. We compute embeddings under three models (TF-IDF, Legal-BERT, and GTE) and develop a validation task built on a hand-curated set of 30 federal circuit splits.

Second, we report that none of the three models meaningfully discriminates between opinions that legally agree and opinions that legally disagree on the same question. All three score within-split opposing opinion pairs at chance level relative to random cross-split pairs. The ordering across models is informative: TF-IDF is most fooled by topic vocabulary overlap (within-split pairs scored 45% higher than random); Legal-BERT comes closest to seeing through the topic confound (gap near zero); GTE is intermediate.

Third, we offer a methodological interpretation: this is not a failure of model selection but a constraint on what text similarity, as a class of methods, can measure. Circuit splits embed an opposition that text similarity is not built to surface. Researchers wishing to study legal disagreement, ideological polarization, or doctrinal evolution using embedding-based similarity should either design analyses that account for this limitation, or augment text similarity with explicit stance-detection methods.

We release our validation set (with full source attribution and selection methodology) and pipeline as a public benchmark for future legal NLP work.

2 Related Work

The most direct antecedent of this work is [Ganguli et al. \[2024\]](#), whose four-step validation pipeline we adopt. Their methodological argument—that model choice is the substantive empirical question, not a technical preliminary—generalizes naturally to any document corpus where the latent structure being measured may not align with what off-the-shelf embeddings capture.

Within legal NLP, [Chalkidis et al. \[2020\]](#) released Legal-BERT, a BERT variant pre-trained on 12GB of English legal text including legislation, court cases, and contracts. Subsequent work has fine-tuned legal embeddings for specific tasks: case retrieval [[Locke and Zuccon, 2018](#)], citation prediction [[Sadeghian et al., 2018](#)], and statute interpretation [[Holzenberger et al., 2020](#)]. Our work occupies a different niche: rather than fine-tuning for a downstream task, we ask what off-the-shelf legal embeddings (including the widely used Legal-BERT checkpoint) actually measure when used as general-purpose similarity functions.

The circuit splits literature in legal scholarship is extensive. [Beim and Rader \[2019\]](#) compile a structured dataset of intercircuit splits from 2005–2013 for use in studies of Supreme Court agenda-setting. [Hellman \[2008\]](#) argues for a stricter standard of what counts as a true circuit split. The annual Congressional Research Service reports on circuit splits [[CRS, 2024, 2025, 2026](#)] provide the primary source from which our validation set is drawn.

A separate line of work uses embedding similarity in legal text without explicit validation. We do not survey this literature exhaustively; our point is methodological. The [Ganguli et al. \[2024\]](#) framework, which we extend here, recommends that any such use be preceded by validation against a task aligned with the downstream research question.

3 Data

3.1 Validation set

The validation set is a manually curated CSV of 30 federal circuit splits spanning 2010–2025 across 10 doctrinal areas. Each row represents one documented circuit split, with named representative cases on each side of the legal disagreement, along with each case’s citation, circuit, year, one-line holding summary, and CourtListener opinion ID.

Sources were drawn primarily from the Congressional Research Service’s annual reports on circuit splits, which catalogue every split that emerged or widened in a given year and remained unresolved as of the report date. We supplemented these with the Seton Hall Circuit Review’s “Current Circuit Splits” feature for earlier years (2010–2022) and used SCOTUSblog “Petitions to Watch” archives for verification.

Three selection criteria were applied. Both sides of the split must have published, precedential opinions with clear majority reasoning; per curiam decisions and memorandum dispositions were excluded. We applied a year filter of 2010 onward to ensure CourtListener coverage and modern citation conventions. Finally, each split had to be expressible as a clear Side A versus Side B disagreement with genuinely opposite outcomes; splits resolving on subtle methodological differences were excluded in favor of substantively opposed holdings.

We aimed for balance, but several structural constraints in the source material make perfect balance impossible. The CRS reports are themselves imbalanced (e.g., the 2025 report contains 29 Criminal Law splits and 1 Antitrust split). We applied a single normalized taxonomy in our `subject_area` field and preserved source labels in notes. We document these limitations in full in the released `data/validation/README.md`.

3.2 Opinion text retrieval

We retrieved full plain-text opinions for all 60 validation cases (30 splits $\times$ 2 sides) using the CourtListener REST API v4. We identified opinions by their CourtListener opinion ID, looked up manually by browsing CourtListener for each case in the validation set, rather than by citation. This choice was forced by an empirical observation: formal F.4th reporter citations are not yet synced for many 2024–2025 cases in CourtListener’s metadata, so a citation-based lookup returns zero results even when the opinion is present in the archive under its numeric ID.

Of the 60 attempted retrievals, 58 succeeded (96.7%). Two cases (split_010 Side A and split_015 Side A) returned 404 errors. These two splits are dropped from validation, leaving 28 evaluable splits. The 56 successfully retrieved opinions generate $\binom{56}{2} = 1,540$ unique unordered pairs, of which 28 are within-split A $\leftrightarrow$ B pairs (one per evaluable split, used as the ground-truth signal) and 1,512 are cross-split pairs (used as the random baseline). The class imbalance (positive rate $28/1,540 \approx 1.8\%$) is comparable in spirit to the patent paper’s interference task ($322/96,580 \approx 0.33\%$), though our sample is two orders of magnitude smaller.

4 Methods

4.1 Four-step pipeline

We adopt the four-step pipeline of [Ganguli et al. \[2024\]](#):

1. **Representation:** each opinion is mapped to a vector in a model-specific embedding space.
2. **Measurement:** cosine similarity is computed for all pairs of opinions in the corpus.
3. **Validation:** each candidate model is scored against a domain-specific ground truth task.
4. **Selection:** the model with the strongest validation performance is selected for downstream analysis.

This paper covers steps 1–3 for the legal domain. Step 4 is left open, because none of our three models passes a meaningful validation threshold, as we discuss in Section 6.

4.2 Models

We evaluate three embedding models, chosen to span the major NLP generations relevant to legal text:

TF-IDF [[Salton et al., 1975](#)] serves as the traditional bag-of-words baseline. We use scikit-learn’s `TfidfVectorizer` with English stopwords, `max_features=10000`, `min_df=2`, and sublinear term frequency scaling.

Legal-BERT [Chalkidis et al., 2020] is a BERT variant pre-trained on 12GB of English legal text including legislation, court cases, and contracts. We use the `nlpaueb/legal-bert-base-uncased` checkpoint from HuggingFace. Opinion text is tokenized to 512 tokens, truncating longer opinions to the first 512, and we apply mean pooling over non-padded token embeddings.

GTE [Li et al., 2023] is the General Text Embeddings model, identified by Ganguli et al. [2024] as the top overall performer in their patent validation. We use the `thenlper/gte-base` checkpoint, with the same tokenization and pooling as Legal-BERT.

We do not evaluate the proprietary OpenAI `text-embedding-3-large` model used in the patent paper, both for full replicability and to keep all candidate models within a single open-weights pipeline.

4.3 Text preprocessing

Each opinion is cleaned via a `preprocess.py` script before embedding. Cleaning steps include: stripping case headers, docket numbers, attorney listings, and court captions; removing standalone disposition boilerplate (AFFIRMED, REVERSED, SO ORDERED); removing inline footnote markers; normalizing whitespace; and repairing UTF-8 encoding artifacts. The aim is to retain only the substantive legal reasoning of the opinion, since headers and dispositions are formulaic and would inflate similarity for purely structural reasons.

4.4 Validation task: two framings

The central methodological choice is how to frame the validation task.

Framework A (topic similarity, patent-paper analog). In Ganguli et al. [2024], the positive label is “these two patents describe the same invention,” and the negative label is “these two patents are randomly paired.” The analog for legal opinions would be: positive = within-split $A \leftrightarrow B$ pair (two opinions adjudicating the same legal question); negative = cross-split pair (two opinions on unrelated questions). A good model under Framework A should score within-split pairs higher than cross-split pairs.

Framework B (stance opposition, our primary framing). Framework A is informative but undemanding. Two opinions in a circuit split necessarily share substantial vocabulary: both cite the same precedents, deploy the same doctrinal categories, and address the same statutory or constitutional provisions. What is unclear is whether a model can see beyond topic to register the opposition in stance.

Framework B operationalizes this question. Under Framework B, the desired behavior is that a model scores within-split $A \leftrightarrow B$ pairs *no higher than* random cross-split pairs. The intuition: because within-split pairs are topically near-identical but conclusionally opposite, a model that distinguishes legal stance from legal topic should not be fooled into rating them similar.

Framework B is the more demanding test and the more informative one for our research question. In our Results we report the gap with its sign and discuss the implications under each framing.

4.5 Validation metrics

For each model m and each evaluable split s , we compute the A $\leftrightarrow$ B similarity $\text{sim}_m(A_s, B_s)$ and a random baseline mean over all cross-split pairs.

We aggregate across splits and report the following summary statistics:

- **Within-split A $\leftrightarrow$ B mean:** average cosine similarity over 28 within-split pairs.
- **Random baseline mean:** average cosine similarity over 1,512 cross-split pairs.
- **Topic-confound gap:** defined as (random mean) – (within-split mean). Under Framework B, a positive gap indicates the model is *not* fooled by topic; a negative gap indicates topic confounding.
- **PR AUC:** precision-recall area under the curve, treating within-split pairs as positives. Chance level is $28/1,540 \approx 0.017$.
- **Discrimination rate:** fraction of splits where the within-split A $\leftrightarrow$ B similarity falls below the median of Side A’s similarity to all other opinions. Values near 0.5 indicate chance performance.
- **Mann-Whitney U test:** two-sided test comparing the distribution of within-split similarities against cross-split similarities. $p > 0.05$ indicates no statistically significant difference.

5 Results

Table 1 reports the full validation results across all metrics.

Table 1: Validation results by model. Gap = random baseline – within-split mean; positive = not fooled by topic (desired under Framework B). PR AUC chance level = 0.017 (28 positive pairs out of 1,540 total). MWU p -values from two-sided Mann-Whitney U test; $p > 0.05$ indicates no statistically significant difference between within-split and cross-split similarity distributions.

Model	Within-split A $\leftrightarrow$ B	Random baseline	Gap	PR AUC	Disc. rate	MWU U	MWU p
TF-IDF	0.125	0.086	−0.039	0.151	0.500	24227	0.556
Legal-BERT	0.913	0.913	+0.0001	0.066	0.464	23933	0.637
GTE	0.832	0.826	−0.007	0.080	0.536	24581	0.465

Figure 1 visualizes the within-split and random-baseline means side by side for each model.

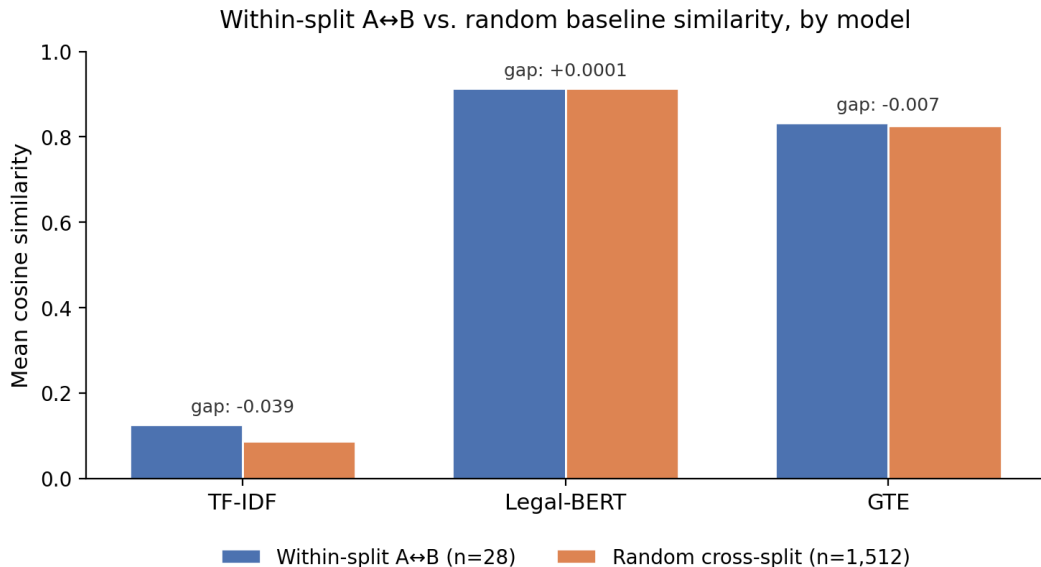


Figure 1: Within-split $A \leftrightarrow B$ vs. random baseline mean cosine similarity, by model. Legal-BERT and GTE absolute magnitudes are inflated by embedding anisotropy [Ethayarajh, 2019]; the meaningful signal is the gap between paired and random distributions, not the absolute level.

Three observations emerge.

Absolute similarity values for transformer models are not directly interpretable. Legal-BERT averages 0.91 across all opinion pairs; GTE averages 0.83. These values reflect a well-documented anisotropy property of BERT-family embeddings [Ethayarajh, 2019], in which sentence embeddings cluster within a narrow cone of the embedding space and pairwise cosines are uniformly inflated. The relevant signal is the gap, not the absolute magnitude.

All three models are at or near zero on the topic-confound gap under Framework B. Legal-BERT’s gap is essentially zero (+0.0001), the cleanest result under Framework B: this model scores within-split pairs no higher than random pairs, indicating it does not over-weight topic vocabulary. GTE’s gap is slightly negative (−0.007), indicating mild topic confounding—within-split pairs score marginally above the random baseline. TF-IDF’s gap is the most negative (−0.039), with within-split pairs scored 45% higher than random pairs on average—the expected behavior for a bag-of-words model that has no representation of meaning beyond word identity.

Discrimination rate hovers around chance for all three models. None of the models can reliably rank a within-split $A \leftrightarrow B$ pair below a random A-to-other pair. GTE marginally exceeds chance (0.536); Legal-BERT marginally falls below (0.464); TF-IDF is exactly at chance (0.500).

PR AUC confirms the near-chance finding. Chance level is 0.017 (28 positive pairs out of 1,540 total). All three models exceed chance, with TF-IDF achieving the highest PR AUC (0.151), consistent with its strong topic-vocabulary signal inflating within-split rankings

under Framework A. Legal-BERT achieves the lowest PR AUC (0.066), consistent with its near-zero gap: a model that does not over-rate within-split pairs will also not rank them particularly high. The two metrics together confirm the central finding: TF-IDF is the most topic-confounded model; Legal-BERT is the least.

Mann-Whitney U test formalizes the near-chance result. All three models return p -values far above the conventional 0.05 threshold (Legal-BERT $p = 0.637$, GTE $p = 0.465$, TF-IDF $p = 0.556$), confirming that no model produces a distribution of within-split similarity scores that is statistically separable from the random baseline.

The two framings yield opposite verdicts. Under Framework A, TF-IDF performs best (within-split similarity 45% above random). Under Framework B, Legal-BERT performs best (gap +0.0001, closest to zero). The two framings yield opposite verdicts on the same numbers, and that opposition is itself a methodological observation: the choice of evaluation framing is, like the choice of model, a substantive decision rather than a technical detail.

6 Discussion

6.1 What this result tells us

The cleanest reading of these results is that the embedding models we evaluated—including a domain-specific legal model—capture the *topic* of a legal opinion but not its *stance* on the legal question. Two opinions in a circuit split discuss the same statute or constitutional provision, marshal the same precedents, and engage the same doctrinal categories. They differ in their disposition: one rules for the plaintiff, the other for the defendant; one reads the statute broadly, the other narrowly. Cosine similarity registers the shared substance and is at best agnostic to the divergent conclusion.

This is not a defect of the models per se. Embedding-based similarity is fundamentally a topic-proximity measure, and it succeeds at that. Our finding is rather that the task of measuring legal agreement requires something that embedding similarity does not provide: a representation of where the opinion comes down, not what it is about.

6.2 Why circuit splits are harder than patent interferences

It is useful to be explicit about why our findings differ from those of [Ganguli et al. \[2024\]](#). The patent paper’s PaECTER model achieves PR AUC of 0.65 on the interference task; GTE achieves 0.64. These are real, large signals: the embedding models do meaningfully separate interfering patent pairs from random pairs.

The structural difference is in the ground truth. A patent interference identifies two documents that describe the same underlying invention. Topic similarity *is* the ground truth signal. A circuit split identifies two documents that adjudicate the same legal question and reach opposite conclusions. Topic similarity is *not* the ground truth signal; it is the confound.

This is a generalizable observation. Validation tasks that target topic identity (interferences, near-duplicate detection, citation prediction) align well with what off-the-shelf embeddings

measure. Validation tasks that target *stance opposition on a shared topic* (circuit splits, disagreement detection, ideological polarization) do not.

6.3 Implications for legal NLP

If text similarity captures topic but not stance, several downstream applications deserve reconsideration:

Precedent retrieval. A model trained to retrieve “similar prior cases” will retrieve cases on the same topic, not cases reaching the same conclusion. For a litigator researching favorable precedent, this distinction is decisive.

Ideological polarization analysis. Studies that operationalize ideological polarization as decreasing similarity between liberal and conservative opinions need to verify that they are measuring stance divergence rather than docket divergence.

Dissent detection. A naive similarity-based dissent flagger will be confounded by the strong topic alignment between majority and dissent within a single case.

6.4 Limitations

Sample size. Our validation set contains 30 splits, of which 28 are evaluable. The patent paper used 322 interference pairs out of 96,580 possible pairs. While our set is small, it is fully hand-curated with documented sourcing.

Topic and circuit imbalance. The D.C. Circuit has zero appearances in our 30 splits; firearms cases are recent-heavy due to post-*Bruen* litigation patterns.

512-token truncation. Both transformer models truncate opinions to their first 512 tokens. A chunk-and-mean-pool approach over the full opinion text would mitigate this.

Two failed retrievals. Two of 30 splits had unfetchable Side A opinions due to CourtListener ID drift. These splits are excluded; the result is robust to their exclusion, but the $n = 28$ sample size is itself a limitation.

Anisotropy. Absolute similarity values for transformer models are inflated by embedding anisotropy and should not be interpreted directly. Within-model comparisons (gap vs. baseline) are valid; across-model comparisons of absolute magnitudes are not.

6.5 Future directions

The most immediately valuable extension is to augment text similarity with explicit stance-detection methods. One option: train a classifier on within-split $A \leftrightarrow B$ pairs (labeled “disagree”) and within-side same-doctrine pairs (labeled “agree”), using model embeddings as features. If even a simple classifier can recover stance from the embedding, this would shift the conclusion from “models do not encode stance” to “models encode stance but not in a way cosine recovers.”

A second extension is the trend analysis left out of this paper: computing within-year average pairwise similarity using the validated model on a larger background corpus of opinions, analogous to the patent paper’s main empirical finding.

7 Conclusion

We extended the validation-based model selection framework of Ganguli et al. [2024] from patent text to legal opinion text and find that the framework reveals a fundamental difference between the two domains. In the patent domain, ground truth (interferences) and embedding similarity are aligned: both measure topic identity. In the legal domain, the most natural ground truth (circuit splits) and embedding similarity are misaligned: the ground truth measures stance disagreement on a shared topic, while embedding similarity measures only topic proximity. Across three models (TF-IDF, Legal-BERT, and GTE), none reliably discriminates between opinions that legally agree and opinions that legally disagree on the same question (Mann-Whitney U, all $p > 0.45$).

We interpret this as a methodological constraint, not a model failure. Researchers using text similarity for substantive legal analysis should be explicit about whether their downstream question requires topic detection, stance detection, or both, and should validate their model choice against a ground truth aligned with their actual research question.

We release our validation set, source attribution, and pipeline as a public benchmark at <https://github.com/kritiat16/tad-judicial-opinions>.

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